NMEC595 - Research Methodology

Lecture 5: Sampling Design

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Lecture Outline

Recap: Research Design

- **Research design** is like a plan that helps you collect and analyze data in the best way.
- **Why it matters**: A good design reduces mistakes, gives trustworthy results, and saves time and resources.
- **Good design**: Should be flexible when exploring new ideas, but strict when describing things clearly.
- **Four main parts**: Sampling, observation, statistics, and how you carry out the research.
- **Type of design**: Should match your research goal—whether you're exploring, describing, diagnosing, or experimenting.
- **Experimental vs. non-experimental**: Choose based on whether you can change (control) the variables or just observe them.
- **Simple vs. formal experiments**: Simple designs are easier, but formal ones give you more control.
- **Good planning**: Is necessary to get results that are accurate and reliable.

Sample Design: Definition and Importance

Definition :

"A sample design is a definite plan for obtaining a sample from a given population. It refers to the technique or procedure the researcher would adopt in selecting items for the sample."

Key Aspects

- **Universe/Population**: The entire collection of objects under study; may be finite or infinite.
- **Sample**: Subset of respondents forming a representative cross-section.
- Sample design is specified prior to data collection.
- Trade-off between precision and constraints of cost and time.

Note: Sample design may also specify sample size (n items from N population).

Seven Steps in Developing Sampling Design

Essential Steps:

1. **Type of universe**: Specify whether the population is finite or infinite.
2. **Sampling unit**: Identify the basic unit—geographical area, structure, social group, or individual.
3. **Source list/Sampling frame**: Compile a comprehensive list of all units in the population (for a finite universe).
4. **Sample size**: Determine the number of units to sample, balancing efficiency, representativeness, and reliability.
5. **Parameters of interest**: Define key estimates such as proportions, means, or subgroup characteristics.
6. **Budgetary constraint**: Assess the financial cost of data collection relative to the risk of incorrect conclusions.
7. **Sampling procedure**: Select the sampling method that minimizes error within resource constraints.

Characteristics of Good Sample Design

Kothari's Five Characteristics

1. **Truly representative**: Accurately reflects the population's features.
2. **Small sampling error**: Limits random variation in estimates.
3. **Viable within budget**: Attains objectives with available resources.
4. **Controlled systematic bias**: Minimizes non-random selection errors.
5. **Reasonable confidence**: Results generalize to the population with statistical assurance.

Optimum Sample: Achieves efficiency, representativeness, reliability, and flexibility.

Two Causes of Incorrect Inferences

Sampling Error:

- Random variations in sample estimates around true parameters
- Compensatory nature; expected value equals zero
- Decreases with sample size increase
- Smaller in homogeneous populations
- Can be measured for given design (*precision*)

Systematic Bias:

- Cannot be reduced by increasing sample size
- More serious than sampling error
- Five major causes (next slide)

Five Causes of Systematic Bias

Sources of Bias:

1. **Inappropriate sampling frame**: The sampling list does not accurately represent the population.
2. **Defective measuring device**: Consistent measurement errors from instruments or questionnaires.
3. **Non-respondents**: Non-response is related to the characteristic being estimated.
4. **Indeterminacy principle**: Subjects alter behavior when aware of being studied.
5. **Natural reporting bias**:
    - Downward bias: Underreporting income for tax.
    - Upward bias: Overreporting income for status.
    - Socially desirable responses in surveys.

Sampling Design: Classification by Representation

1. Probability Sampling

- Every population element has a known, nonzero probability of selection.
- Selection uses random methods, reducing bias.
- Enables valid statistical inference and error estimation.
- *Examples*: Simple random, stratified, systematic sampling.

2. Non-Probability Sampling

- Probability of selection is not known or cannot be calculated.
- Selection is non-random, often based on convenience or judgment.
- More prone to selection bias; statistical inference is limited.
- *Examples*: Convenience, judgment, quota sampling.

Sampling Design: Classification by Element Selection

1. **Unrestricted Sampling**
    - Each element is selected independently from the population.
    - No structural constraints on the sampling process.
    - *Example*: Simple random sampling.
2. **Restricted Sampling**
    - Selection process follows specific rules or structures.
    - Ensures representation of subgroups or utilizes groupings.
    - *Examples*: Stratified sampling, cluster sampling.

Simple Random Sampling: Definition and Methods

Definition

- In simple random sampling, n elements are selected from a population of size N in such a way that every possible combination of n elements is equally likely.
- There are ${}^{N}C_n$ unique samples, and each has probability $\frac{1}{{}^{N}C_n}$ of being chosen.

How Sampling Is Done

- **Physical randomization**: Methods like drawing lots, shuffling cards, or using a lottery system.
- **Random number methods**: Use of published random number tables (Tippett, Fisher, Yates) or computer-generated random numbers to select elements from the population list.

Statistical Principle

- **Law of Statistical Regularity**: If the sample is drawn randomly and is large enough, the sample tends to have characteristics similar to those of the population.

Simple Random Sampling: Properties and Example

Illustrative Example

- Suppose $N = 6$ elements: $\{a, b, c, d, e, f\}$.
- Select a sample of $n = 3$ elements.
- There are 20 possible samples ($^6C_3 = 20$), such as abc, abd, …, def.
- Each sample has probability $1/20$ of being chosen.

Systematic Sampling: Definition and Procedure

Definition

- Systematic sampling is a probability sampling method that selects elements from an ordered population at regular intervals.
- The sampling interval k is calculated as $k = \frac{N}{n}$, where N is the population size and n is the desired sample size.

Procedure

- Randomly select a starting point r from among the first k elements.
- Select every k-th element thereafter: positions r, $r + k$, $r + 2k$, $\ldots$, continuing until the sample contains n elements or the list ends.
- Only the initial selection is random; subsequent selections are determined by the interval k.

Systematic Sampling: Properties and Example

Properties

- Process is simpler and faster than repeated random draws.
- Ensures uniform coverage across the population list.
- Risk of bias if the ordered list has periodic patterns that coincide with the sampling interval.
- Works best when the population list is randomly ordered.

Illustrative Example

- Population size $N = 1000$; desired sample size $n = 40$.
- Calculate $k = \frac{1000}{40} = 25$.
- Randomly select r between 1 and 25.
- Select elements at positions r, $r + 25$, $r + 50$, $\ldots$, up to the end of the list.

Stratified Random Sampling: Concept and Procedure

Definition

- Stratified random sampling is a probability sampling method in which the population is divided into distinct, non-overlapping subgroups called *strata*.
- Each stratum is internally homogeneous with respect to a chosen characteristic, while different strata differ from one another.

Sampling Procedure

- Partition the population into strata based on known attributes such as age, gender, region, or income level.
- Apply simple random sampling independently within each stratum.
- Combine the samples drawn from all strata to form the final sample.

Stratified Random Sampling: Properties and Motivation

Key Properties

- Reduces sampling variability by controlling variation within each stratum.
- Produces more precise population estimates than simple random sampling when the population is heterogeneous.
- Guarantees representation of all important subgroups.
- Prevents over-representation or under-representation of specific groups.

Motivation

- Stratification uses prior knowledge of population structure.
- When subgroup characteristics differ substantially, stratified sampling improves efficiency and reliability of statistical estimates.

Stratified Sampling: Allocation Methods I

Deciding Sample Size for Each Stratum

- **Proportional Allocation**
  - Assigns sample sizes in proportion to the size of each stratum in the population.
  - Formula: $n_i = n \times P_i$, where $P_i = \frac{N_i}{N}$, N_i is the size of stratum i, N is total population, n is total sample size.
  - *Example*: If Stratum A contains 30% of the population, then 30% of the sample will be drawn from Stratum A.

- **Optimum Allocation**
  - Takes into account both the size and the variability (standard deviation) of each stratum.
  - Formula: $n_i = n \times \frac{N_i \sigma_i}{\sum N_k \sigma_k}$, where σ_i is the standard deviation within stratum i.
  - More samples are allocated to strata that are larger or more variable, to increase efficiency and precision.

Stratified Sampling: Allocation Methods II

Cost-Optimal Allocation

- **Definition**
  - Allocates sample sizes considering stratum size, variability, and cost per observation in each stratum.
  - Formula: $n_i = n \times \frac{N_i \sigma_i / \sqrt{C_i}}{\sum N_k \sigma_k / \sqrt{C_k}}$, where C_i is the unit cost in stratum i.
- **Explanation**
  - Fewer samples are assigned to strata where sampling is more expensive, balancing statistical efficiency and budget constraints.
  - Used when costs of data collection differ between strata.
- *Summary*
  - The allocation method should be chosen based on study goals, the variability in strata, and available resources.

Cluster Sampling: Concept and Procedure

Definition

- Cluster sampling is a probability sampling method in which the population is divided into groups called *clusters*, often using natural or geographic boundaries.
- Clusters may represent regions, institutions, households, or other groupings, and are typically heterogeneous within themselves.

Sampling Procedure

- Divide the entire population into non-overlapping clusters.
- Randomly select a predetermined number of clusters.
- Either sample all units within chosen clusters (*one-stage*) or sample a subset of units from each selected cluster (*two-stage* or multistage).

Cluster Sampling: Features and Applications

Types and Applications

- **Area Sampling**: Clusters are defined by geographic areas such as districts, towns, or villages, making it useful for large-scale field surveys.

Advantages

- Highly cost-effective, especially when the population is widely dispersed geographically.
- Reduces travel, administrative, and listing costs for researchers.

Disadvantages

- Provides less precise estimates than simple random sampling because units within the same cluster tend to be similar (intra-cluster correlation).
- Each sampled unit yields less information about the population compared to truly random samples.

Probability Sampling: Multi-Stage Sampling (Overview)

What is Multi-Stage Sampling?

- Multi-stage sampling collects data through a sequence of sampling steps.
- Each stage involves selecting groups (units) from a list (sampling frame).
- The process continues through multiple levels, ending with the final units of study (often households or individuals).

Difference from Cluster Sampling

- In one-stage cluster sampling, clusters are chosen and all units within each selected cluster are studied.
- In multi-stage sampling, additional sampling is performed within each selected cluster or group, sometimes over several stages.

Probability Sampling: Multi-Stage Sampling (Why and How)

Why Use Multi-Stage Sampling?

- A complete list of all individuals may not be available.
- Lists often exist only for higher-level units (e.g., states, districts, villages).
- The method reduces cost and fieldwork by narrowing the area step by step.

General Procedure

1. Stage 1: Randomly select primary units from a list (e.g., states).
2. Stage 2: Within each selected primary unit, randomly select secondary units (e.g., districts).
3. Stage 3: Within each selected secondary unit, randomly select final units (e.g., households).

Probability Sampling: Multi-Stage Sampling (Probability and Example)

Selection Probability Formula

- If m stages are used and the same number of units are selected at each stage,

$$p = \prod_{j=1}^{m} \frac{n_j}{N_j}$$

- Here, n_j is the number chosen at stage j, N_j is the total at stage j.

Numerical Example

- Stage 1: $N_1 = 10$ states, select $n_1 = 2$
- Stage 2: in each state, $N_2 = 6$ districts, select $n_2 = 3$
- Stage 3: in each district, $N_3 = 100$ households, select $n_3 = 10$
- Probability for a household:

$$p = \frac{2}{10} \times \frac{3}{6} \times \frac{10}{100} = 0.01$$

Non-Probability Sampling: Features and Types

Key Characteristic

- In non-probability sampling, not every population member has a known or equal chance of being selected.
- There is no statistical basis for estimating the probability of selection.
- As a result, sampling error and margin of error cannot be measured.

Main Types

1. **Purposive (Judgement/Deliberate) Sampling**
   - The researcher selects units based on their own judgment about which are most appropriate.
   - This can introduce personal bias, but may be useful in exploratory or specialized studies.
2. **Quota Sampling**
   - The population is divided into groups (quotas) by characteristics such as age or gender.
   - Interviewers select required numbers from each group at their discretion, not randomly.

Non-Probability Sampling: Convenience Sampling and Limitations

Convenience (Haphazard) Sampling

- Samples are chosen based on what is easiest or most accessible to the researcher.
- This method is the quickest and least expensive, but it is highly prone to bias.
- Often used for pilot studies or when rapid results are needed, but not suitable for generalization.

Limitations of Non-Probability Sampling

- Greater risk of selection bias compared to probability sampling.
- Results are less reliable for making inferences about the entire population.
- Used mainly in exploratory research, case studies, or situations where random sampling is impractical.

Factors Determining Sample Size

Key Considerations :

1. **Desired precision**: Margin of error acceptable
2. **Confidence level**: 95% or 99% typical
3. **Population variance (σ^2)**: Larger variance needs larger sample
4. **Population size (N)**: Limits maximum sample size
5. **Parameters of interest**: Proportions, means, sub-group estimates
6. **Sampling method**: Some methods need larger samples
7. **Budgetary constraint**: Cost per unit $\times$ sample size
8. **Time availability**: Data collection timeline
9. **Nature of population**: Homogeneous vs. heterogeneous

Principle: Sample size should be optimum—neither excessively large nor too small.

Selecting Appropriate Sampling Procedure

Seven Selection Criteria :

1. **Nature of universe**: Homogeneous or heterogeneous?
2. **Availability of sampling frame**: Complete population list?
3. **Type of inquiry**: Exploratory, descriptive, or hypothesis-testing?
4. **Desired precision**: Required accuracy level?
5. **Cost and time**: Budget and timeline constraints?
6. **Expertise available**: Researcher competence and training?
7. **Bias control**: Which method minimizes systematic bias?

Two Costs in Sampling:

- Cost of collecting data
- Cost of incorrect inference from data

Summary: Key Concepts

1. **Sample design** is definite plan determined *before* data collection
2. **Census** impractical for most research; **samples** more efficient
3. **Seven steps**: Universe type, sampling unit, frame, size, parameters, budget, procedure
4. **Good sample**: Representative, small error, viable cost, controlled bias, reasonable confidence
5. **Two error types**: Sampling error (measurable, decreases with n) vs. Systematic bias (5 causes)
6. **Probability sampling**: Random selection, known probability, allows statistical inference
7. **Non-probability**: No error estimation; useful for exploratory studies
8. **Selection depends on**: Nature, frame, inquiry type, precision, cost, time, expertise

Probability Sampling Comparison

Method	**Key Feature**	**Best For**
Simple Random	Equal probability $1/{}^{N}C_n$	Homogeneous populations
Systematic	Every k-th item	Large lists, random order
Stratified	Homogeneous strata	Heterogeneous populations
Cluster/Area	Geographic groups	Dispersed populations, cost saving
Multi-stage	Sequential clustering	Large-scale surveys

Practical Guidelines

Researcher Must:

- Select design with *smaller sampling error* for given size and cost
- Control *systematic bias* through proper procedures
- Ensure sampling frame is *comprehensive, correct, reliable*
- Remember: Large sample increases cost *and* may enhance bias
- Better design > larger sample for improving precision
- Apply *Law of Statistical Regularity*: Random sample reflects universe

Quote : *"In practice, people prefer a less precise design because it is easier to adopt and systematic bias can be controlled in a better way."*

Questions and Discussion

Critical Questions

"Is my sample truly representative?"
"Have I minimized both sampling error and bias?"
"Is my sampling procedure optimal for the cost?"
"Can I estimate sampling error with my design?"

Thank you!